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## 8.9 Van't Hoff Equation

This equation gives the quantitative temperature dependence of equilibrium constant (K). The relation between standard free energy change ($\Delta G^\circ$) and equilibrium constant is

$$\Delta G^\circ = -RT\ln K \qquad (1)$$

We know that

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \qquad (2)$$

Substituting (1) in equation (2)

$$-RT\ln K = \Delta H^\circ - T\Delta S^\circ$$

Rearranging

$$\ln K = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \qquad (3)$$

Differentiating equation (3) with respect to temperature,

$$\frac{d(\ln K)}{dT} = \frac{\Delta H^\circ}{RT^2} \qquad (4)$$

Equation 4 is known as differential form of van't Hoff equation.

On integrating the equation 4, between $T_1$ and $T_2$ with their respective equilibrium constants $K_1$ and $K_2$.

$$\int_{K_1}^{K_2} d(\ln K) = \frac{\Delta H^\circ}{R}\int_{T_1}^{T_2}\frac{dT}{T^2}$$

$$[\ln K]_{K_1}^{K_2} = \frac{\Delta H^\circ}{R}\left[-\frac{1}{T}\right]_{T_1}^{T_2}$$

$$\ln K_2 - \ln K_1 = \frac{\Delta H^\circ}{R}\left[-\frac{1}{T_2} + \frac{1}{T_1}\right]$$

$$\ln \frac{K_2}{K_1} = \frac{\Delta H^\circ}{R}\left[\frac{T_2 - T_1}{T_2T_1}\right]$$

$$\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303\,R}\left[\frac{T_2 - T_1}{T_2T_1}\right] \qquad \ldots\ldots\ldots (5)$$

Equation 5 is known as integrated form of van't Hoff equation.

**Problem:**

For an equilibrium reaction $K_p = 0.0260$ at $25^\circ C$ $\Delta H = 32.4\ kJmol^{-1}$, calculate $K_p$ at $37^\circ C$

**Solution :**

$T_1 = 25 + 273 = 298\ K$

$T_2 = 37 + 273 = 310\ K$

$\Delta H = 32.4\ KJmol^{-1} = 32400\ Jmol^{-1}$

$R = 8.314\ JK^{-1}\,mol^{-1}$

$K_{P1} = 0.0260$

$K_{p2} = ?$

$$\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303\,R}\left[\frac{T_2 - T_1}{T_2T_1}\right]$$

$$\log \frac{K_2}{K_1} = \frac{32400}{2.303 \times 8.314}\left(\frac{310 - 298}{310 \times 298}\right)$$

$$= \frac{32400 \times 12}{2.303 \times 8.314 \times 310 \times 298}$$

$$= 0.2198$$

$$\frac{K_2}{K_1} = \text{anti}\log 0.2198 = 1.6588$$

$$K_2 = 1.6588 \times 0.026 = 0.0431$$

### Evaluate Yourself

5) The equilibrium constant for the following reaction is 0.15 at 298 K and 1 atm pressure.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g);$$

$$\Delta H_f^\circ = 57.32\ KJmol^{-1}$$

The reaction conditions are altered as follows.

a) The reaction temperature is altered to $100^\circ$ C keeping the pressure at 1 atm, Calculate the equilibrium constant.